

20250505EB_MN (RNA, NCT, ATP)

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2025年5月5日 星期一

Use 2.5mM T8 full supp (made by no glucose ADMEM/F12 and Neurobasal-A) for differentiation and experiment.

Cells still not that optimal

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2025年5月7日 星期三

T2 -> T3

2025年5月9日 星期五

T3 -> T4 (~14 mL for over-weekend)

2025年5月12日 星期一

T4 -> T5

2025年5月14日 星期三

T5 -> T6

2025年5月16日 星期五

T6 -> T6 (~14mL for over-weekend)

2025年5月19日 星期一

T6 -> T7

Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.

6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

cell info ^

	A	B	C
1	cell	passage	well#
2	WT11	16+7	6 & 5

Table1 ^

	A	B
1	run1	volume (mL)
2	trypsin	5
3	HI-FBS/CTWM	10

Table3 ✓

	A	B	C	D	E	F	G	H	I	J
1		conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	measurement 1	2.29	100	5	11.45	A	6	1200	6	2824
3	measurement 2	2.55	99	5	12.75	B, C	24	239	22	2176
4		2.42								

Plate A ^

	1	2	3
A			
B			

Plate B						
	1	2	3	4	5	6
A			111		92	
B						
C			110			
D						



Plate C (for Sydney)						
	1	2	3	4	5	6
A						
B						
C						
D						



2025年5月20日 星期二

Change T7 -> T7 if added virus

2025年5月21日 星期三

T7-> T7 if not added virus

2025年5月22日 星期四

T7 -> T8

2025年5月24日 星期六

1/2 T8 -> 1/2 T8

2025年5月27日 星期二

1/2 T8 -> 1/2 T8

2025年5月28日 星期三

Treat Plate A
Prepare MG132:
stock: 10.5mM
1uL 10.5mM in 210uL --> 50uM

60uL 50uM in 2mL --> 1.5uM

Table2				
	A	B	C	D
1	Condition		treat date	treat time
2	A	2.5mM gluc 1h	5/29	11:00
3	B	0.15mM gluc 1h	5/29	
4	C	2.5mM gluc 24h	5/28	12:10
5	D	0.15mM gluc 24h	5/28	
6	E	2.5mM gluc w/ DMSO 24h	5/28	12:08
7	F	2.5mM gluc w/ 1.5uM MG132 24h	5/28	

Well1			
	1	2	3
A	A	C	E
B	B	D	F

2025年5月29日 星期四

1/2 T8 -> 1/2 T8

RNA extraction Plate A

treat cells at 10:50 am

collect cells at 11:50 am

Imaging Plate B

Plate B conditions						
	1	2	3	4	5	6
A			1	2	A	H
B			4	3	B	G
C			1	2	C	F
D			4	3	D	E

vLH92

Pre-treatment: 3 loops, 10min interval

Post-treatment: 7 loops, 10 min interval

prepare media with different conc. of glucose1							
	A	B	C	D	E	F	G
1		diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)	media volume
2	A	1100	20	55	2000	36.4	1963.6
3	B	20	7.5	2.6666666667	1000	375.0	625.0
4	C	20	2.5	8	1600	200.0	1400.0
5	D	2.5	0.5	5	1000	200.0	800.0
6	E	2.5	0.15	16.6666666667	1000	60.0	940.0
7	F	2.5	0.05	50	1000	20.0	980.0
8	G	2.5	0.015	166.6666666667	1000	6.0	994.0
9	H		0				

O.C. , 40X, zoom1.5, 600x900, 16-bit				
	A	B	C	D
1	channel	bit	power (%)	exposure time
2	488	16	20	300
3	Dapi-ex-Green-em	16	20	500

Timelapse Imaging

Pretreatment: 3 loops
Post-treatment: 7 loops

vLH111,110

media preparation_SirDNA							
	A	B	C	D	E	F	G
1	condition #	glucose conc. (mM)	volume	FCCP (1000x)	2CG (333x)	SirDNA dilution	SirDNA volume
2	1	2.5	2000			5000	0.4
3	2	0.15	2000			5000	0.4
4	3	0	2000			5000	0.4
5	4	0	2000	2	6	5000	0.4

Table5						
	A	B	C	D	E	F
1			110		111	
2	number	condition	treatment time	image 1h	treatment time	image 1h
3	1	2.5mM	4:55	5:55	2:50	3:52
4	2	0.15mM	5:40	6:38	3:20	4:22
5	3	0mM	6:25	7:25	3:50	4:55
6	4	0mM+FCCP+2DG	7:00	7:58	4:20	5:20

O.C. , 60X, 600x900, 12-bit					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	12	30	300 ms	
3	640	12	10	100 ms	
4	488	12	50	1s	

10 positions/well

110: pre-activation: 3 loops, 30s interval; activation: 30 loops, 30s interval

111: pre-acitvation: 3 loops, 30s interval; activation: **25 loops**, 30s interval